NARAYANA
IIT ACADEMY
INDIA

45 YEARS OF EXCELLENCE

Sec: SR_*CO-SC(MODEL-A) Date:30-06-24
Time: 3 Hrs Max. Marks: 300

30-06-24_SR.IIT_STAR CO-SC(MODEL-A)_JEE MAIN_CTM-13_SYLLABUS

PHYSICS: 100% CUMULATIVE

Syllabus covered From 22-04-23 TO 29-06-24

CHEMISTRY: 100% CUMULATIVE

Syllabus covered From 22-04-23 TO 29-06-24

MATHEMATICS: 100% CUMULATIVE

Syllabus covered From 22-04-23 TO 29-06-24

PHYSICS MAX.MARKS: 100

SECTION – I

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A body is projected up from the surface of the earth with a velocity equal to $\frac{3}{4}th$ of its escape velocity. If R be the radius of the earth, the height it reaches is

A) $\frac{3R}{10}$ B) $\frac{9R}{7}$ C) $\frac{8R}{5}$ D) $\frac{9R}{5}$

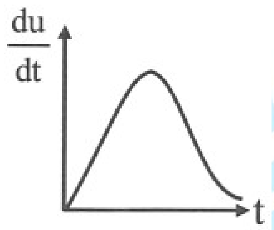
2. The equation of SHM of a particle is given as $2\frac{d^2x}{dt^2}+32x=0$ where x is the displacement from the mean position. The period of its oscillation (in seconds) is

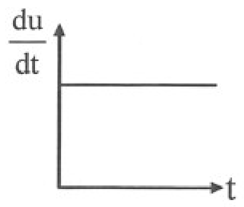
A) 4 B) $\frac{\pi}{2}$ C) $\frac{\pi}{2\sqrt{2}}$ D) 2π

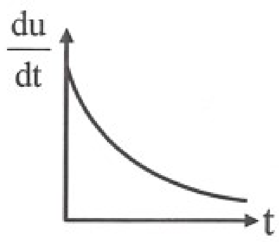
3. A long straight wire carrying a current of 30A is placed in an external uniform magnetic field of induction $4\times10^{-4}T$. The magnetic field is acting parallel to the direction of current. The magnitude of the resultant magnetic induction in tesla at a point 2.0 cm away from the wire is

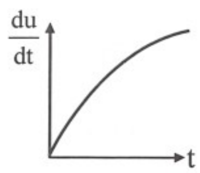
A) 10^{-4} B) 3×10^{-4} C) 5×10^{-4} D) 6×10^{-4}

4. Rate of increment of energy in an inductor with time in series LR circuit getting charge with battery of e.m.f. E is best represented by: [inductor has initially zero current]

A) 

B) 

C) 

D) 

5. In series LCR circuit voltage drop across resistance is 8 volt, across capacitor is 12 volt and across inductor is 6 volt. Then:

A) Voltage of the source will be leading current in the circuit

B) Voltage drop across each element will be less than the applied voltage

C) power factor of circuit will be 4/3

D) none of these

6. Main scale of a vernier caliper has 100 divisions in 5 cm. Its vernier scale has 25 divisions in one cm. The least count is

A) 0.01 cm B) 0.005 cm C) 0.01 mm D) none of these

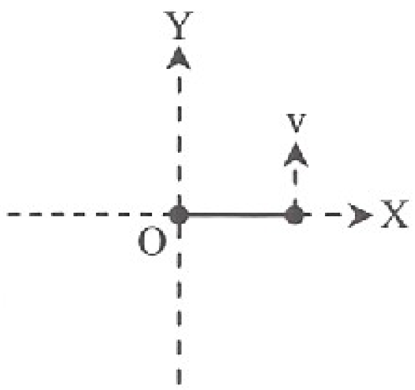
7. The frequency of fundamental tone in an open organ pipe of length 0.48 m is 320 Hz. Speed of sound is 320 m/sec. Frequency of fundamental tone in closed organ pipe of the same length will be

A) 153.8 Hz B) 160.0 Hz C) 320.0 Hz D) 143.2 Hz

8. A box containing N molecules of a perfect gas at temperature T_1 and pressure P_1. The number of molecules in the box is doubled keeping the total kinetic energy of the gas same as before. If the new pressure is P_2 and temperature T_2, then

A) $P_2 = P_1, T_2 = T_1$ B) $P_2 = P_1, T_2 = \frac{T_1}{2}$ C) $P_2 = 2P_1, T_2 = T_1$ D) $P_2 = 2P_1, T_2 = \frac{T_1}{2}$

9. A small sphere of mass m and carrying a charge q is attached to one end of an insulating thread of length a, the other end of which is fixed at (0, 0) as shown in figure. There exists a uniform electric field $\vec{E} = -\vec{E}_0\,\hat{j}$ in the region. The minimum velocity which should be given to the sphere at (a, 0) in the direction shown so that it is able to complete the circle around the origin is (There is no gravity)



A) $\sqrt{\frac{5qE_0a}{m}}$ B) $\sqrt{\frac{3qE_0a}{m}}$ C) $\sqrt{\frac{qE_0a}{m}}$ D) $2\sqrt{\frac{qE_0a}{m}}$

10. A magnet is suspended horizontally in the earth's magnetic field. When it is displaced and released, it oscillates in a horizontal plane with a period T. If a piece of wood of same M.I. as the magnet is attached to the wood, the new period of oscillation of the system would be

A) $\frac{T}{3}$ B) $\frac{T}{2}$ C) $\frac{T}{\sqrt{2}}$ D) $\sqrt{2}\,T$

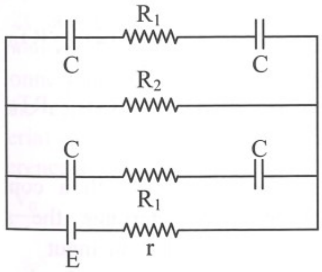
11. 10 g of ice at 0^0C is mixed with 100g of water at 50^0C. What is the resultant temperature of mixture

A) 31.2^0C B) 32.8^0C C) 36.7^0C D) 38.2^0C

12. When a 100 W, 240 V bulb is operated at 200 volt, the current in it is

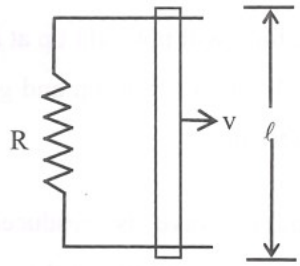
A) 0.35 A B) 0.42 A C) 0.50 A D) 0.58 A

13. In figure $E=5V, r=1\,\Omega, R_1=1\,\Omega, R_2=4\Omega, \& C=3\mu F$. The numerical value of charge on each plate of capacitor is (during steady state)



1) $3\ \mu C$ B) $6\ \mu C$ C) $12\ \mu C$ D) $24\ \mu C$

14. A conducting rod of resistance r moves uniformly with a constant speed v. If the rod keeps moving uniformly, then the amount of force required is



A) $\frac{vB^2\ell^2}{R}$ B) $\frac{2vB^2\ell^2}{(R+r)}$ C) $\frac{vB^2\ell^2}{(R+r)}$ D) zero

15. One mole of an ideal monoatomic gas at temperature T_0 expands slowly according to the law P/V = constant. If he final temperature is $2T_0$, heat supplied to the gas is

A) $2RT_0$ B) $\frac{3}{2}RT_0$ C) RT_0 D) $\frac{1}{2}RT_0$

16. A wooden block of mass 8 kg is tied to a string attached to the bottom of a tank. The block is completely inside the water. Relative density of wood is 0.8. Taking $g=10\ m/s^2$, what is the tension in the string ?

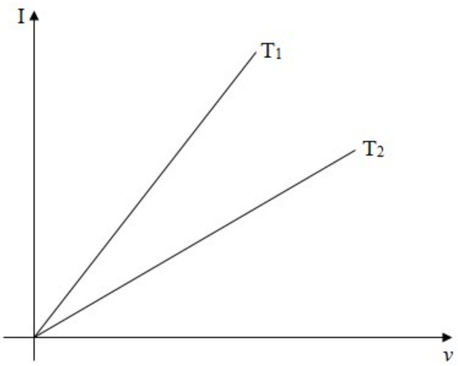
A) 100 N B) 80 N C) 50 N D) 20 N

17. Bulk modulus of elasticity for isobaric process is

A) equal to that of isochoric process B) equal to that of isothermal process

C) zero D) infinite

18. The current I and voltage V graphs for a given metallic wire at two different temperature are shown in fig. Then we conclude

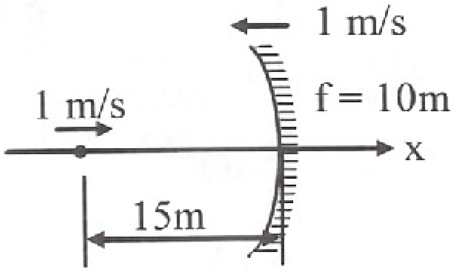


A) $T_1>T_2$ B) $T_1<T_2$ C) $T_1=T_2$ D) $T_1=2T_2$

19. The equation for a wave traveling in x-direction on a string is $y=(3\,cm)\sin\left[\left(\pi\,cm^{-1}\right)x-(314)s^{-1}t\right]$. Then find acceleration of a particle at $x=6\,cm$ at $t=0.11\,\text{sec}$

A) $2952\,cm/s^2$ B) $5904\,m/s^2$ C) $2952\,m/s^2$ D) zero

20. A point object moves in + x-direction with v = 1 m/s along the principal axis of the concave mirror of focal length $f=10\,m$. When the mirror moves with a velocity $\vec{V}_m=-\hat{i}\,m/s$ and the object is at a distance of p = 15 m, the speed of the image is



A) $-8\hat{i}\,m/s$ B) $-9\hat{i}\,m/s$ C) $-6\hat{i}\,m/s$ D) none of these

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

21. An aeroplane was flying horizontally with a velocity of 720 km/h at an altitude of 490 m. When it is just vertically above the target a bomb is dropped from it. How far horizontally it missed the target (in m) ?

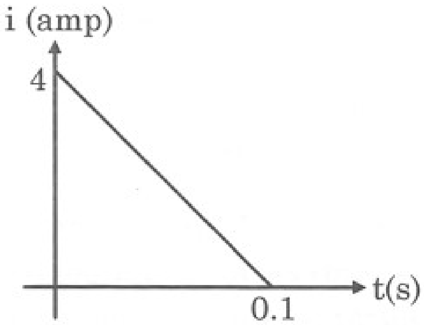
22. A wheel is subjected to uniform angular acceleration about its axis. Initially its angular velocity is zero. In the first 2 sec, it rotates through an angle θ_1; in the next 2 sec, it rotates through an additional angle θ_2. The ratio of θ_2/θ_1 is ?

23. A 30 cm long cylinder floats vertically in mercury at 0^0C. If the temperature rises to 100^0C, the increase in length of the cylinder (in cm) under mercury in x find the value of $100\,x$. The density of mercury at $0^0C=13.6\,g/cm^3$, density of iron at $0^0C=7.6\,g/cm^3$, coefficient of volume expansion of mercury $=1.82\times10^{-4}/^0C$ and coefficient of cubical expansion of iron $=3.51\times10^{-5}/^0C$.

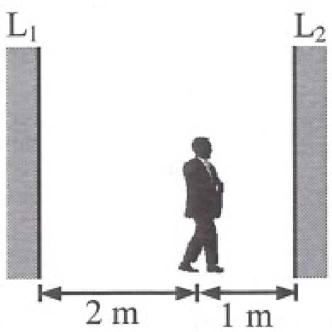
24. In copper, each copper atom releases one electron. If a current of 1.1 A is flowing in the copper wire of uniform cross-sectional area of diameter 1 mm, drift velocity (in mm) of electrons (Density of copper $=9\times10^3\,kg/m^3$, Atomic weight of copper = 63) is $x\,mm$. Find the value of $100\,x$.

25. A current of 3 A is flowing in a plane circular coil of radius 4 cm and number of turns 20. The coil is placed in a uniform magnetic field of magnetic induction 0.5 T. Then the dipole moment (in Am^2) of the coil is x. Find the value of $10\,x$?

26. Some magnetic flux is changed from a coil of resistance 10 ohm. As a result an induced current is developed in it, which varies with time as shown in figure. The magnitude of change in flux through the coil in Webers is



27. Two mirror labeled L_1 for left mirror and L_2 for right mirror in the figure are parallel to each other and 3 m apart. A person standing 1 m from the right mirror (L_2) looks into this mirror and sec a series of images. Find the second nearest image seen in the right mirror is situated at a distance. If it is n meter from person. Find value of n.



28. A steel rod of length 25 cm has a cross-sectiona area of $0.8\,cm^2$. The force in (N) required to stretch this rod by the same amount as the expansion produced by heating it through 10^0C is coefficient of linear expansion of steel is $10^{-5}\,{}^0C^{-1}$. Young's modulus of steel is $2\times10^{10}\,N/m^2$.

29. In a certain double slit experimental arrangement, interference fringes of width 1.0 mm each are observed when light of wavelength $5000\,A^0$ is used. Keeping the set-up unaltered if the source is replaced by another of wavelength $6000\,A^0$, the fringe width will be (in mm) x. Then the value of $10x$.

30. A telescope has an objective lens of 10 cm diameter and is situated at a distance of one kilometer from two objects. The minimum distance between these two objects, which can be resolved by the telescope, when the mean wavelength of light is $5000\,A^0$, of the order of (in mm)

CHEMISTRY MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

31. Iron form a sulphide with approximate formula, Fe_7S_8. Iron exists in both +2 and +3 oxidation states. The ratio of Fe^{+2} atoms to Fe^{+3} atoms is ?(Average Oxidation state of sulphur is -2)

A) 3:2 B) 2:3 C) 2:5 D) 5:2

32. In which of the following the order is not correct, according to the property indicated against it?

A) $Na > Mg > Al$ - Second ionization potential

B) P < S <Cl -non metallic character

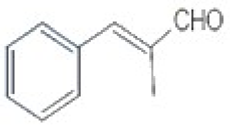
C) $F_2 > Cl_2 > Br_2 > I_2$- Oxidising power

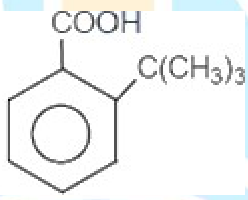
D) $P_2O_5 < SO_3 < Cl_2O_7$- Acidic nature

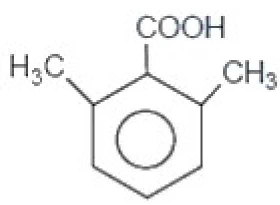
33. The equilibrium: $P_4(g)+6Cl_2(g) \rightleftharpoons 4PCl_3(g)$ is obtained by mixing equal moles of P_4 and Cl_2 in an evacuated vessel. Then at equilibrium

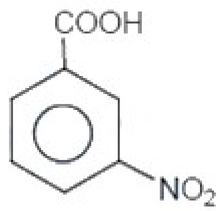
A) $[Cl_2]>[PCl_3]$ 2) $[Cl_2]>[P_4]$ 3) $[P_4]>[Cl_2]$ 4) $[PCl_3]>[P_4]$

34. Acidic strength is more in case of

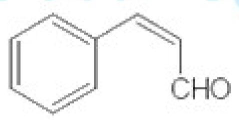
A) 

B) 

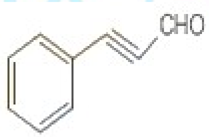
C) 

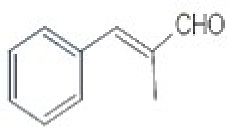
D) 

35. An aldehyde, (P) which does not undergo self aldol condensation gives benzaldehyde and two moles of (Q) on ozonolysis. Compound (Q), on oxidation with silver ions gives oxalic acid. The structure of (P) is given as

A) 

B) 

C) 

D) 

36. Among the following complexes (K to P)

$K: K_3[Fe(CN)_6], L:[Co(NH_3)_6]Cl_3, M: Na_3[Co(oxalate)_3], N:[Ni(H_2O)_6]Cl_2$

$O: K_2[Pt(CN)_4], and\ P:[Zn(H_2O)_6](NO_3)_2$

The diamagnetic complexes are

A) K,L,M,N B) K,M,O,P C) L,M,O,P D) L,M,N,O

37. Which of the following statements is false

A) Two different solutions of sucrose of same molality prepared in different solvent will have the

same depression in freezing point

B) The osmotic pressure of a solution is given by the equation $\pi = CRT$ (where. C is the molarity of

non volatile non electrolyte solute)

C) Decreasing order of osmotic pressure for 0.01 M aqueous solutions of barium chloride, potassium

chloride, acetic acid and sucrose is $BaCl_2 > KCl > CH_3COOH > sucrose$

D) According to Raoult'slaw, the vapour pressure exerted by a volatile component of a solution is

directly proportional to its mole fraction in the solution

38. How many revolutions an electron will make in 2nd shell of He^+ in one second?

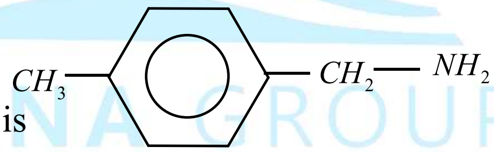
A) 6.66×10^{15} B) 3.33×10^{15} C) 2.22×10^{15} D) 1.11×10^{15}

39. Gabriel-Phthalimide synthesis is suitable for the preparation of

A) $CH_3-CH_2NHCH_2CH_3$

B) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-NH_2$

C) 

D) 

40. The incorrect statement among the following is

A) Amylopectin is water soluble

B) Lactose is composed of $\beta-$glucose and $\beta-$galactose and joined by 1,4 – glycosidic linkage

C) Sucrose is a non-reducing sugar

D) Glycine is optically inactive amino acid

41. 600 ml O_2 in a closed container is heated to convert 15% O_2 into O_3. The final volume of gaseous mixture is ___ $ml\left(3O_2 \rightarrow 2O_3\right)$

A) 510 B) 60 C) 570 D) 90

42. A 0.10 M solution of fluoride ions is gradually added to a solution containing $Ba^{2+}.Ca^{2-}.and\ Pb^{2+}ions.$ each at a concentration of 1 x 10^{-3} M. In what order, from first to last, will the precipitates of BaF_2. CaF_2 and PbF_2 form?

	K_{sp}
BaF_2	1.8×10^{-7}
CaF_2	1.5×10^{-10}
PbF_2	7.1×10^{-7}

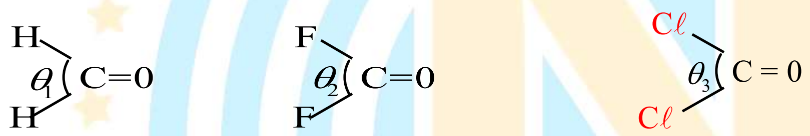
A) CaF_2,PbF_2.BaF_2 B) BaF_2,CaF_2.PbF_2

C) PbF_2,BaF_2.CaF_2 D) CaF_2,BaF_2.PbF_2

43. 0.5 mole each of two ideal gas $A\left(c_v=\frac{3}{2}R\right) and\ B\left(c_v=\frac{5}{2}R\right)$ are taken in a container and expanded reversibly and adiabatically from V = 1 litre to V = 4 litre startely from initial temperature T = 300K, ΔH for the process (in cal/mol) is

A) -500cal B) -900cal C) -450cal D) -600cal

44.

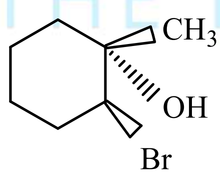


Identify the correct order of bond angle

A) $\theta_1>\theta_2>\theta_3$ B) $\theta_1>\theta_3>\theta_2$ C) $\theta_3>\theta_2>\theta_1$ D) $\theta_1=\theta_2=\theta_3$

45.

; True statement regarding 'X'

A) 

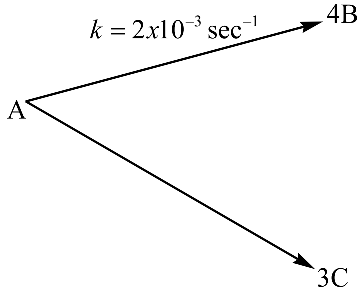
B) The reaction is regioselective

C) NBS donate Br^+ in this reaction D) All of these

46. The emf of a standard cadmium cell is 1.01832 V at 298 K. The temperature coefficient of the cell is $-5.0\times10^5 VK^{-1}$ ΔH for the cell reaction is (approx.)

A) $-196.508 KJmol^{-1}$ B) $-98.28 KJmol^{-1}$ C) $-199.38 KJmol^{-1}$ D) $-99.69 KJmol^{-1}$

47. For the following parallel chain reaction



What will be that value of overall half-life of A in minutes? $\left[\text{Given that } \frac{[B]_t}{[C]_t} = \frac{16}{9}\right]$

A) 3.3 B) 6.3 C) 3.6 D) None

48.

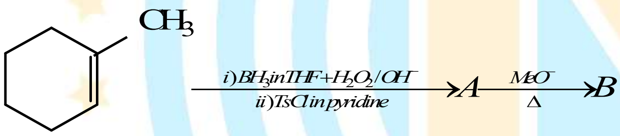
$$\underset{(A)}{CH_3-\underset{\substack{|\\CH_3}}{CH}-\underset{\substack{|\\Br}}{CH}-CH_3} \xrightarrow[\text{Elimination}]{\text{Hoffman}} (B) \xrightarrow{HBr} (C)$$

$$(B) \xrightarrow{HBr/\text{peroxide}} (D)$$

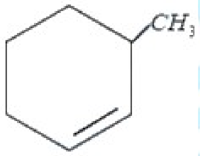
Correct order of rate of S_N1 for A,C& D will be :-

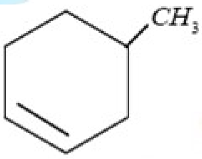
A) A > C > D B) C > D > A C) A = C > D D) C > A > D

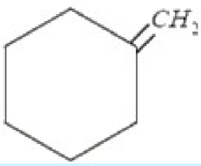
49.

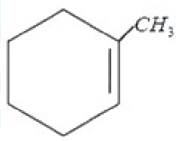


The product 'B' is

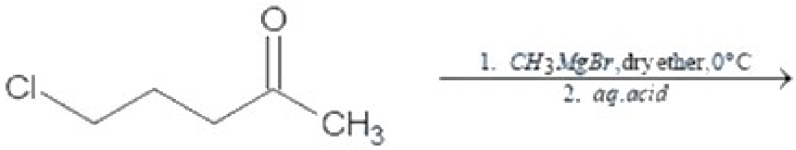
A) 

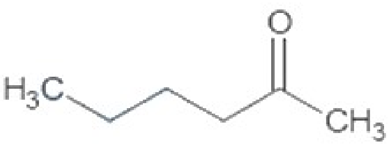
B) 

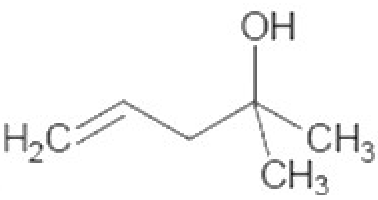
C) 

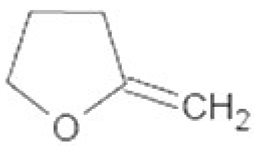
D) 

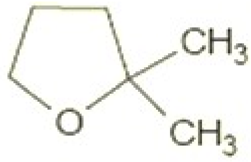
50. The major product in the following reaction is



A) 

B) 

C) 

D) 

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:
Marking scheme: +4 for correct answer, -1 in all other cases.

51. The ion A^{n+} is oxidized to AO_3^- by MnO_4^- changing to Mn^{2+} in acidic medium. Given that 2.68×10^{-3} mole of A^{n+} requires 1.61×10^{-3} mole of MnO_4^-. Thus, n is

52. Among the following, the total number of reactions/ processes in which the entropy increases are:

a) $2NaHCO_3(s) \rightarrow Na_2CO_3(s) + CO_2(g) + H_2O(g)$

b) A liquid crystalline into a solid.

c) Temperature of crystalline solid is raised from zero K to 100K.

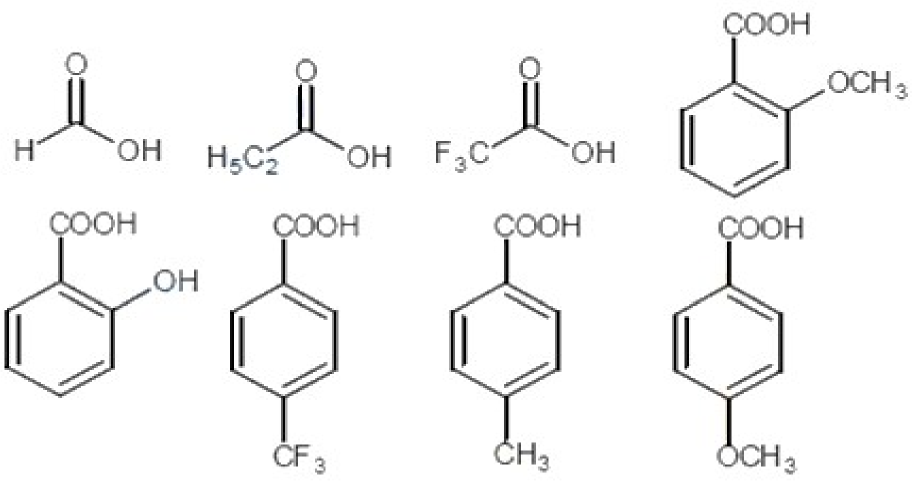
d) Hard boiling of an egg.

e) Intermixing of gases at constant temperature

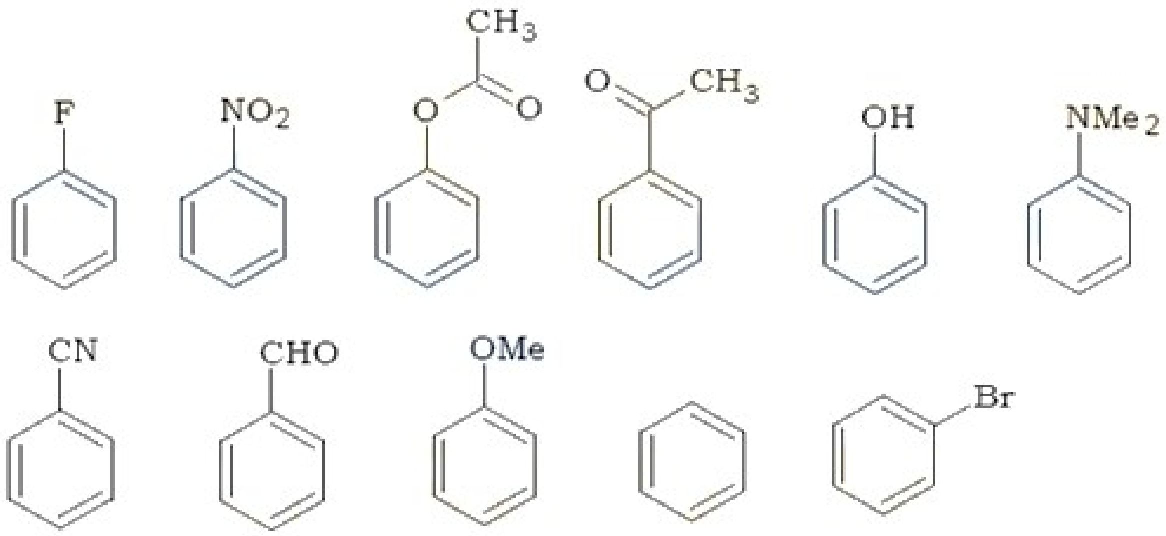
f) Boiling of water

53. A hydrogen atom in its ground state is irradiated by light of wavelength 972.5A^0 and the ground state energy of hydrogen atom as -13.6 ev, the number of lines present in the emission spectrum is ___ .

54. How many of the following acids are more acidic than benzoic acid?



55. How many of the following compounds are more reactive than chlorobenzene towards nitration?



56. If K_{sp} of $PbSO_4$ is 4 x 10^{-10}, then what is the loss in wt. of $PbSO_4$ if it is washed with 5 lit of water is __mg(mol.wt of $PbSO_4$ = 303 g/mol)

57. Resistance of an aqueous solution containing 2 mole NH_4Cl and is filled in between two electrodes which are 20 cm apart was found to be 100 ohm. Calculate the $\wedge_m \left(Scm^2mol^{-1}\right) for\ NH_4Cl(aq.)$

58. How many out of following reagents will change 1-cyclohexylmethanol into cyclohexane carbaldehyde

i) Pyridinium chloro chromate ii) H^+/ $KmnO_4$ iii) Tollen's reagent

iv) NBS/ Δ v) Red hot Cu/300^0C vi) CrO_3 in aq. H_2SO_4& acetone

vii) $LiAlH_4$ viii) $NaBH_4$

59. How many of the following compounds are polar? $CH_3Cl, NH_3, CO_2, H_2O, CH_4, PCl_5, C_2H_2$;

60. The total number of resonating structures in aAniliniumion $\left(C_6H_5\overset{\oplus}{N}H_3\right)$?

MATHEMATICS MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

61. $\lim_{n\to\infty}\sum_{r=1}^{n}\cot^{-1}\left(r^2+\frac{3}{4}\right)=$

A) $\tan^{-1}(2)$ B) $\frac{\pi}{4}$ C) $\frac{\pi}{2}$ D) $\tan^{-1}(3)$

62. Let $f(x)=x^3\left\{\sqrt{x^2+\sqrt{x^4+1}}-x\sqrt{2}\right\}$. Then $\lim_{x\to\infty} f(x)$ is equal to

A) $\frac{1}{2\sqrt{2}}$ B) $\frac{1}{4\sqrt{2}}$ C) $\frac{3}{4\sqrt{2}}$ D) does not exist

63. Let $f: R\to R$ defined as $f(x)=x+\sin x$, then value of $\int_0^{2\pi} f^{-1}(x)dx=$

A) $2\pi^2$ B) $2\pi^2-2$ C) $2\pi^2+2$ D) π^2

64. If $9+f^{11}(x)+f^{1}(x)=x^2+f^2(x)$ be the differentialequation of a curve and let P be the point of minima of this curve then the number of tangents which can be drawn from P to the circle $x^2+y^2=8$ is

A) 2 B) 1 C)0 D) 3

65. Let f:[-3 1] $\to R$ be given as $f(x)=\begin{cases}\min\left\{(x+6),x^2\right\} & -3\le x<0\\ \max\left\{\sqrt{x},x^2\right\} & 0\le x\le 1\end{cases}$ If the area bounded by y=f(x) and x-axis is A, then the value of 12 A=____

A)40 B) 46 C)82 D)39

66. $\frac{1}{1!.100!}+\frac{1}{3!.98!}+\frac{1}{5!.96!}+\text{........}+\frac{1}{(101)!}=$

A) $\frac{2^{100}}{100!}$ B) $\frac{2^{101}}{101!}$ C) $\frac{2^{100}}{(101)!}$ D) $\frac{2^{51}}{(101)!}$

67. Let A be 3×3 Invertible matrix $|Adj(24A)|=|Adj(3Adj(2A))|$ then absolute value of $|A|$ is ($|A|$ is determinant of A)

A) 8 B) 64 C) 4 D)2

68. Range of the function $f(x)=\cos^{-1}\left(-\{x\}\right)$ (where $\{.\}$ is fractional part function) is

A) $\left[\frac{\pi}{2},\pi\right)$ B) $\left(\frac{\pi}{2},\pi\right]$ C) $\left(\frac{\pi}{2},\pi\right)$ D) $\left(0,\frac{\pi}{2}\right]$

69. If $y^{\frac{1}{4}}=x+\sqrt{x^2+1}$, $\left(1+x^2\right)y_2+xy_1=ky$ then k is equal to (Where y_n is nth derivative of y)

A) 16 B) 8 C)4 D) 1

70. $f(x)=\begin{cases}2-\left|x^2+5x+6\right| & x\neq -2\\ a^2+1 & x=-2\end{cases}$ Then the range of a so that f(x) has a maxima at x=-2 is

A) $[-1,1]$ B) $(-\infty,-1]\cup[1,\infty)$ C) $[-2,2]$ D) $(-\infty,1]$

71. Let $A=\{1,2,3,........,22\}$, **B** is a subset of **A** having exactly 11 elements. The sum of elements of all possible subsets of B is

A) $253\times{}^{21}C_{10}$ B) $252\times{}^{21}C_{11}$ C) $230\times{}^{21}C_{11}$ D) $253\times{}^{21}C_9$

72. If $z_1,z_2,z_3\in C$ satisfy the system of the equation given by $|z_1|=|z_2|=|z_3|=1$, $z_1+z_2+z_3=1$ and $z_1z_2z_3=1$ such that $Im(z_1)<Im(z_2)<Im(z_3)$ then the value of $\left|z_1+z_2^{\,2}+z_3^{\,3}\right|^2$ is equal to

A) 0 B) 5 C) 4 D) 1

73. Let α,β be the roots of the equation $x^2-2x+2=0$ and $\left(\frac{\alpha}{\beta}\right)^n=1,(n\in N)$ then the least value of **n** is equal to

A) 3 B) 2 C) 4 D) 5

74. Let $\begin{vmatrix}1+x & x & x^2\\ x & 1+x & x^2\\ x^2 & x & 1+x\end{vmatrix}=ax^5+bx^4+cx^3+dx^2+ex+f$ then match the elements of column-I with column-II

Column-I	**Column-II**
a) Value of f is	p) 0
b) Value of e is	q) 1
c) Value of a+c is	r)-1
d) Value of b+d is	s) 3

A) a-q, b-s, c-r, d-q B) a-s, b-q, c-r, d-q

C) a-s, b-s, c-q, d-r D) a-r, b-r, c-s, d-q

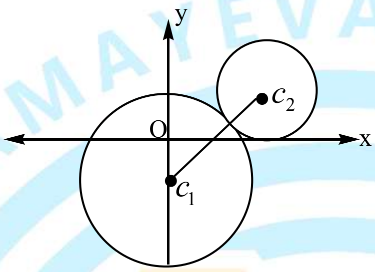
75. Let the first term 'a' and the common ratio 'r' of a geometric progression be positive integers. If the sum of its squares of first three terms is 21, then the sum of first 10 terms of the given G.P is equal to

A) 1023 B) 511 C) 2047 D)241

76. If the term independent of x in the expansion of $\left(\sqrt{\frac{x}{3}}+\frac{3}{2x^2}\right)^{10}$ *is* $\frac{m}{n}$ (m, n are relatively prime) then the value of (m+n) is

A) 9 B)4 C)5 D) 10

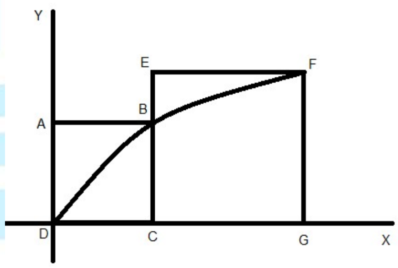
77. In the given figure, the equation of the larger circle is $x^2+y^2+4y-5=0$ and the distance between centres is 4. Then the equation of the smaller circle is:



A) $\left(x-\sqrt{7}\right)^2+(y-1)^2=1$ B) $(x-1)^2+\left(y-\sqrt{7}\right)^2=1$

C) $x^2+y^2=2\sqrt{7}x+2y$ D) $\left(x-\sqrt{7}\right)^2+(y-2)^2=4$

78. ABCD and EFGC are squares and the curve $y=k\sqrt{x}$ passes through the origin D and the points B and F. The ratio $\frac{FG}{BC}$ is equal to



A) $\frac{\sqrt{5}+1}{2}$ B) $\frac{\sqrt{3}+1}{2}$ C) $\frac{\sqrt{5}+1}{4}$ D) $\frac{\sqrt{3}+1}{4}$

79. Let $f(x)=2x(2-x), 0\le x\le 2$,then the number of real roots of the equation $f\left(f(f(x))\right)=\frac{x}{2}$ is equal to

A) 8 B)16 C) 9 D)10

80. If a tangent of slope 2 of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is normal to the circle $x^2+y^2+4x+1=0$, then the maximum value of 'ab' is equal to

A) 16 B) 8 C)4 D) $\sqrt{5}$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

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81. If the lines $\frac{x}{1}=\frac{y}{2}=\frac{z}{3},\frac{x-1}{3}=\frac{y-2}{-1}=\frac{z-3}{4}$ and $\frac{x+k}{3}=\frac{y-1}{2}=\frac{z-2}{h}$ are concurrent, then the value of 2k+h is equal to

82. Let $a_1,a_2,a_3,.....,a_n$ be real numbers in arithmetic progression such that $a_1=15$ *and* a_2 is an integer. Given $\sum_{r=1}^{10}(a_r)^2=1185$. If $\sum_{r=1}^{n}(a_r)$ and maximum value of n is N for which $S_n \geq S_{(n-1)}$, then the value of $N-10$ is equal to

83. The polynomial R(x) is the remainder obtained dividing x^{2007} by x^2-5x+6. If R(0) can be expressed as $ab(a^c-b^c)$, then the value of $\left[\frac{a+b}{20}+c\right]$ is equal to (Where [.] is greatest integer function)

84. If $\underset{n\to\infty}{Lt}\sum_{r=1}^{4n}\frac{\sqrt{n}}{\sqrt{r}\left(3\sqrt{r}+4\sqrt{n}\right)^2}=\lambda$, then $10\lambda=$

85. The number of pairs ordered (x,y) satisfying the equation $\sin x+\sin y=\sin(x+y)$ and $|x|+|y|=1$ is equal to

86. The range of values of 'k' for which the equation $2\cos^4 x-\sin^4 x+k=0$ has atleast one solution is $[\lambda,\mu]$. Then the value of $(9\mu+\lambda)$ is equal to

87. Let $\bar{a},\bar{b},\bar{c}$ be three unit vectors such that $\bar{a}$ is perpendicular to the plane of $\bar{b}$ and $\bar{c}$ If the angle between $\bar{b}$ and $\bar{c}$ is $\frac{\pi}{3}$ then $\left|\bar{a}\times\bar{b}-\bar{a}\times\bar{c}\right|^2$ is equal to

88. A curve $g(x)=\int x^{27}\left(1+x+x^2\right)^6\left(6x^2+5x+4\right)dx$ is passing through (0, 0), then $\frac{7}{3^6}g(1)$ is equal to

89. If α,β are the roots of $x^2+x+2=0$ and γ,δ are the roots of $x^2+3x+4=0$ then $(\alpha+\gamma)(\alpha+\delta)(\beta+\gamma)(\beta+\delta)$ is

90. If $A=\begin{bmatrix} a & a^2-1 & -2 \\ a+1 & 1 & a^2+4 \\ -2 & 4a & 5 \end{bmatrix}$ is symmetric, then a is equal to